

THE DOPAMINERGIC VOICE: COMPARING ACOUSTIC PROSODY & VENTRAL STRIATAL ACTIVATION AS BIOMARKERS FOR ANHEDONIA

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Background

- **Depression** is the leading cause of disability worldwide, affecting hundreds of millions
- **Anhedonia** (inability to feel pleasure) is the most treatment-resistant symptom
- **fMRI**, the current gold standard, costs \$500+/session

Research Question: Using ClinicalWhisper, can a 15-min voice recording detect anhedonia as accurately as a \$500 fMRI scan?

One Pathway, Two Readouts

Anhedonia is driven by **reduced dopamine** in the mesolimbic pathway, which produces two independently measurable effects:

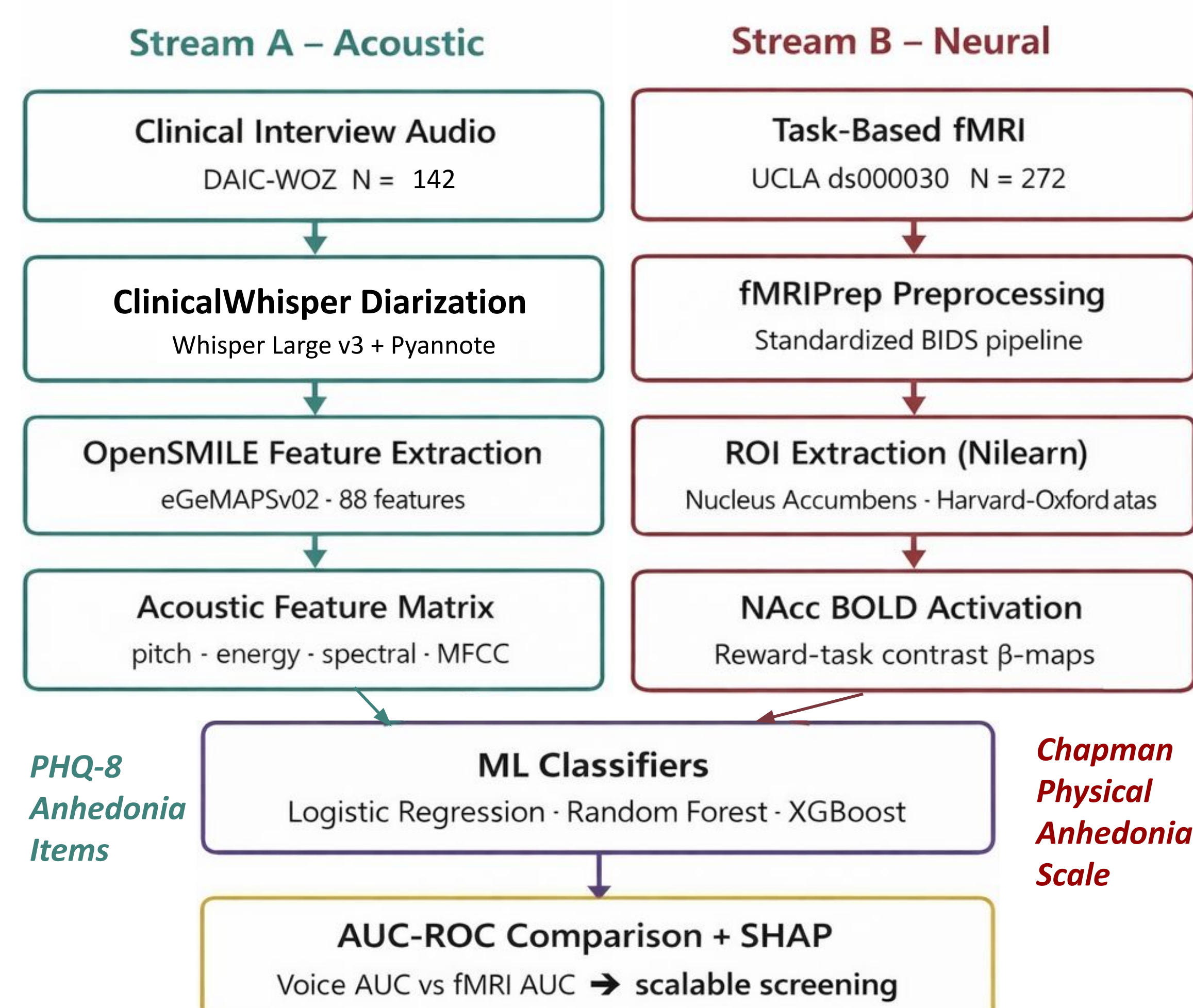
Brain reward circuit changes - \$\$\$

Voice & speech changes - \$

VTA → Nucleus Accumbens → Cortex

Hypothesis: Voice features can serve as a low-cost proxy for neural reward signals with the same biology, but different measurement window.

Methods - Two-Stream Study Design



Results

Metric	Stream A — Acoustic	Stream B — Neural
Dataset	DAIC-WOZ (N = 142)	ds000030 (UCLA) (N = 234)
Best AUC (pre-reg)	0.612 ± 0.131	0.580 ± 0.038
Best AUC (exploratory)	0.651 ± 0.066	-
Permutation p	p = 0.075 / 0.026	p = 0.057

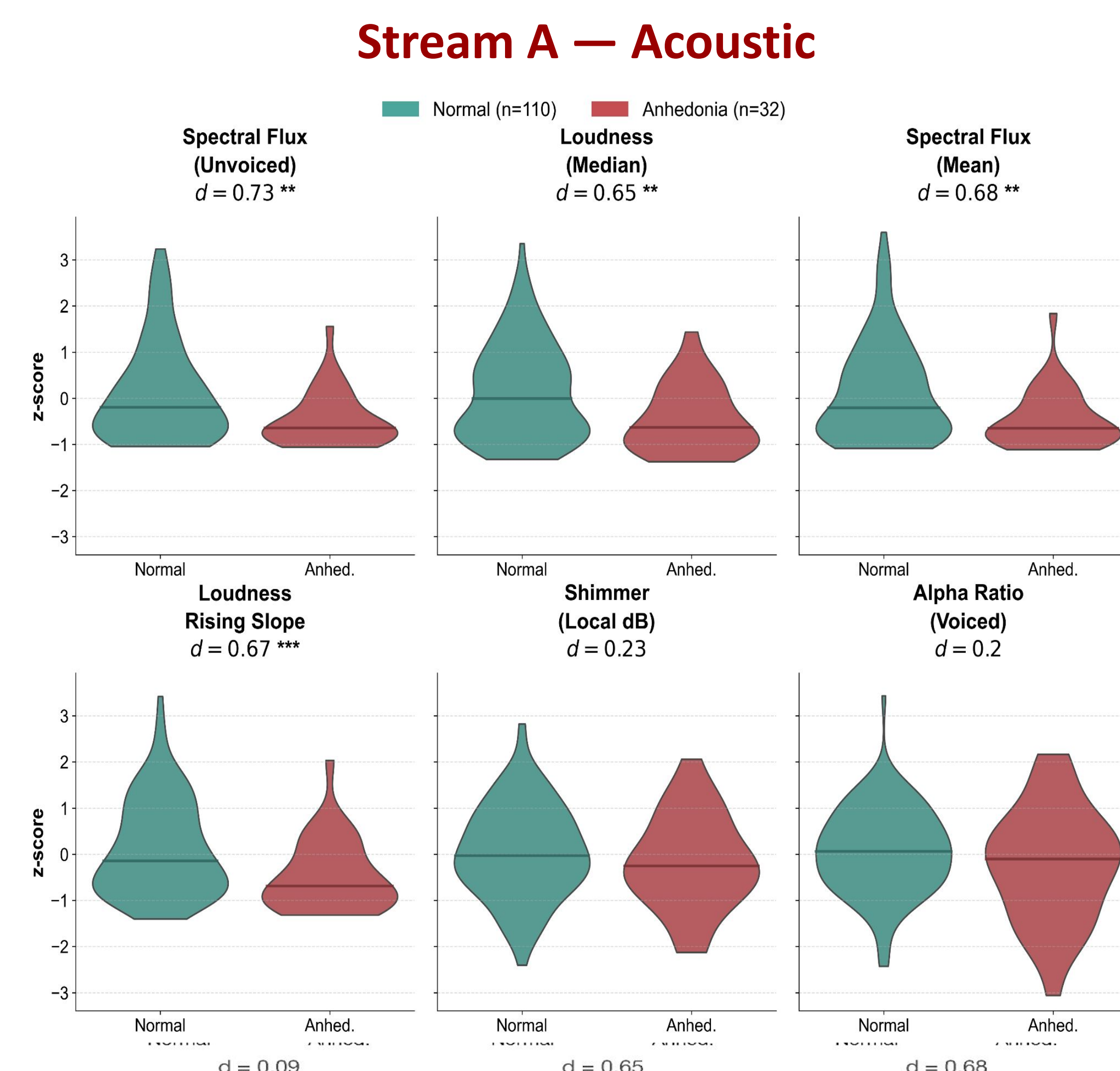


Fig 1. Anhedonic speakers show flatter, quieter voices. Loudness ($d=0.65$) and spectral flux ($d=0.68$) show medium-to-large separation ($N=142$).

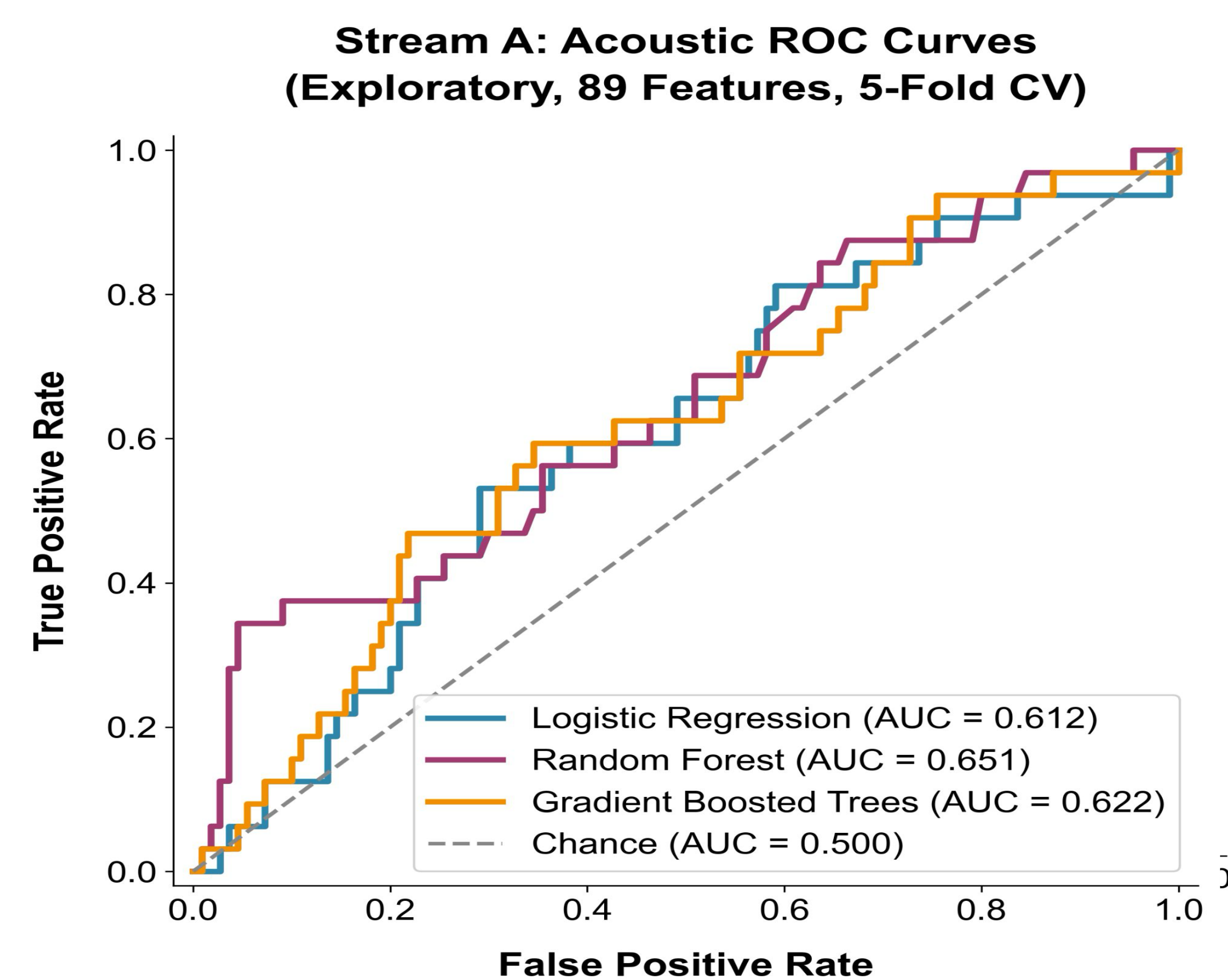


Fig 2. Voice classifies anhedonia above chance; RF achieves best AUC (0.651). Full interview, 5-fold CV, $N=142$.

Stream B — Neural Activation

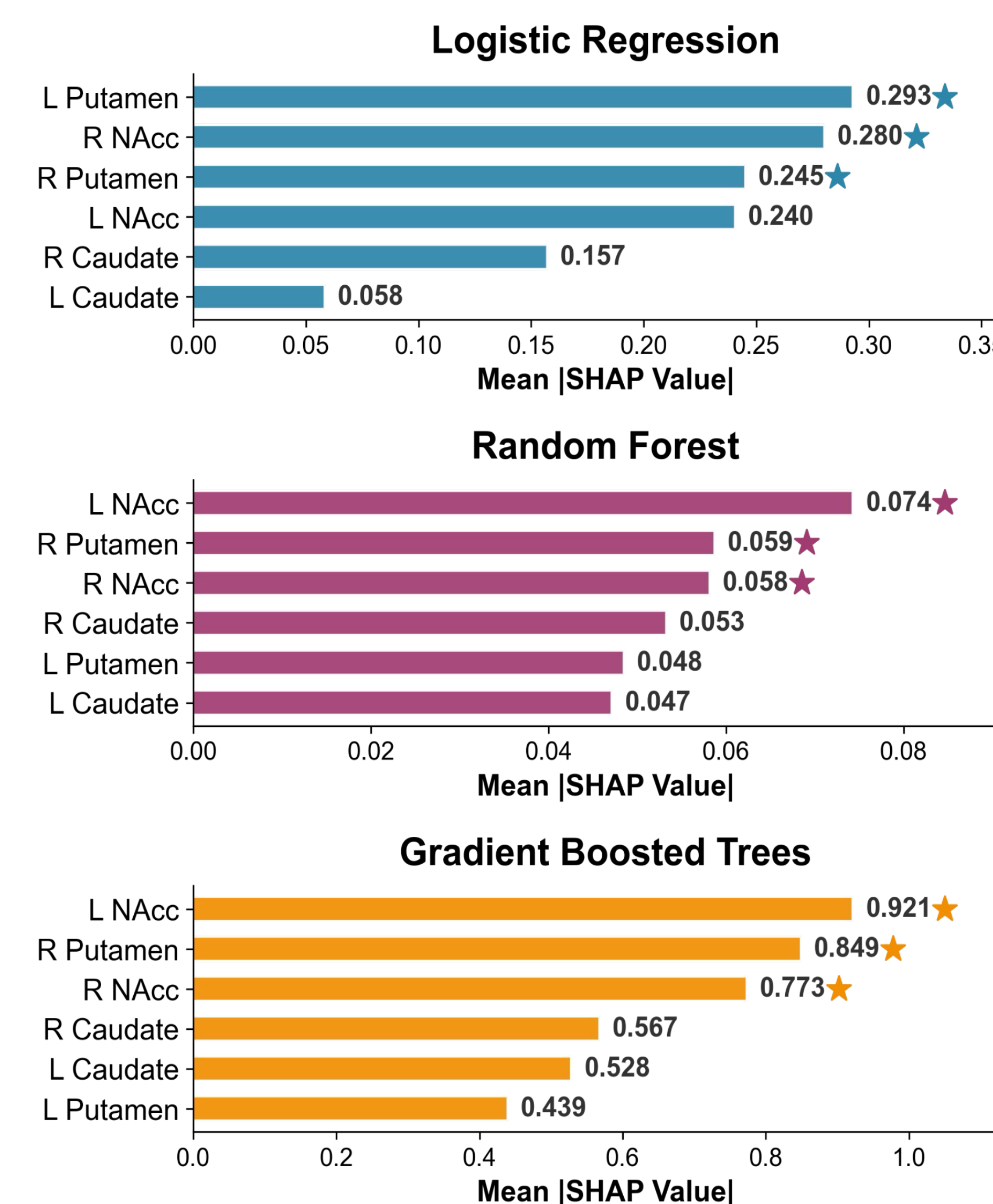


Fig 4. L Putamen and R NAcc — mesolimbic dopamine nodes; drive classification across all models ($N=234$).

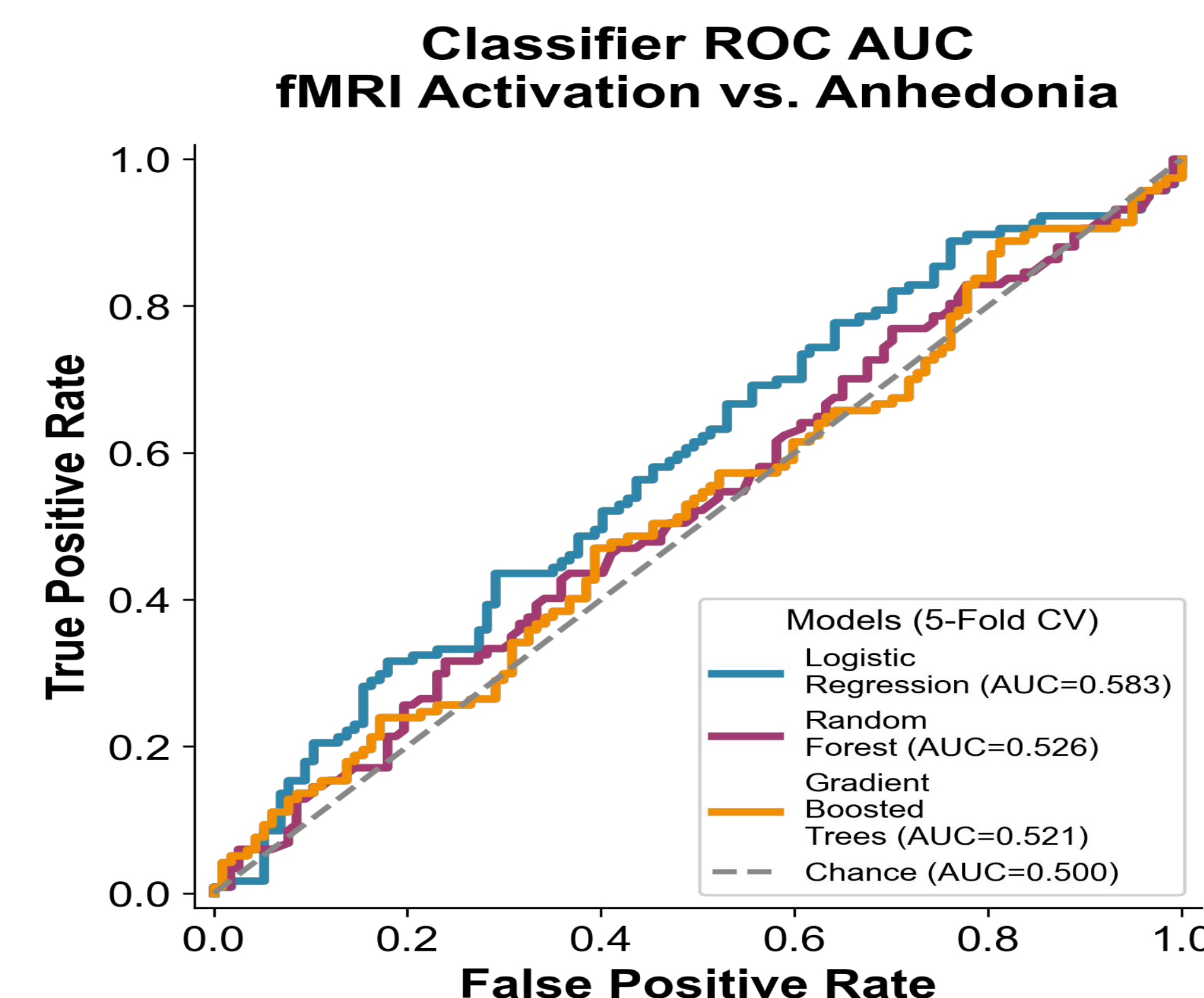


Fig 5. Striatal activation independently classifies anhedonia (LogReg AUC=0.580, $p=0.057$), converging with the acoustic stream.

Key Findings

- **Stream A (full interview, exploratory): AUC = 0.651, $p = 0.026$**
- **Stream A (diarized, pre-registered): AUC = 0.612, $p = 0.075$**
- **Stream B (fMRI, $n=234$): AUC = 0.580, $p = 0.057$**
- **Voice approaches fMRI-level anhedonia classification**
- **Diarrization reveals interviewer accommodation as a contributor to the acoustic signal**

Conclusions

- Voice-based classification matches fMRI-level performance ($\Delta AUC = +0.032$), supporting acoustic prosody as a low-cost anhedonia proxy.
- Diarrization reveals interviewer accommodation as a classifiable signal source ($\Delta AUC = -0.039$) — a novel methodological finding.
- Both streams show above-chance trends (Stream A: $p = 0.075$; Stream B: $p = 0.057$), consistent with a shared mesolimbic pathway signal that approaches conventional significance.

Limitations & Future Work

- Independent samples (DAIC-WOZ vs. ds000030) — mitigated by age-matched subsetting
- Diarrization reveals partial interviewer accommodation confound in classification
- Future: within-subject multimodal design, larger clinical samples, pre-registered replication with isolated speech

References & Acknowledgments

Data Sources: DAIC-WOZ (USC ICT);

UCLA ds000030 (OpenNeuro)

• All code is publicly available via ClinicalWhisper (github.com/czhou732/ClinicalWhisper)

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Open Data & Code
osf.io/4d6ey